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Weightlifting system

Abstract

An exercise system includes a base structure including a first cross brace including first cross brace mounting plates disposed at each end of the first cross brace. A pair of base guide rail supports is secured to and extended substantially orthogonal from the first cross brace and includes a pair of base guide rail receivers. A pair of base pulley supports is secured to and extended substantially orthogonal from the first cross brace and further includes a pulley-support pulley housing. A pair of pulley-support pulleys is rotatably secured to the pulley-support pulley housing. A second cross brace includes a central portion substantially parallel to the first cross brace. The central portion is secured to the pair of base guide rail supports and the pair of base pulley supports. Two end portions are disposed at an angle to the central portion and terminate at second cross brace mounting plates disposed at each end of the two end portions and two central-portion pulley housings. A central-portion pulley is rotatably secured to each of the two central-portion pulley housings.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4339125	12/1981	Uyeda	N/A	N/A
4898381	12/1989	Gordon	N/A	N/A
4907798	12/1989	Burchatz	N/A	N/A
4974838	12/1989	Sollenberger	N/A	N/A
5011139	12/1990	Towley, III	N/A	N/A
5094449	12/1991	Stearns	N/A	N/A

5298004	12/1993	Davis	N/A	N/A
5328429	12/1993	Potash	482/99	A63B 21/0058
5374229	12/1993	Sencil	N/A	N/A
5476428	12/1994	Potash	482/99	A63B 21/063
5665036	12/1996	Hsieh	N/A	N/A
5688216	12/1996	Mauriello	N/A	N/A
5954619	12/1998	Petrone	N/A	N/A
5971897	12/1998	Olson	N/A	N/A
D428949	12/1999	Simonson	N/A	N/A
6102835	12/1999	Webber	N/A	N/A
D437370	12/2000	Webber	N/A	N/A
D438267	12/2000	Webber	N/A	N/A
6217483	12/2000	Kallassy	N/A	N/A
6220988	12/2000	Williams	N/A	N/A
6261205	12/2000	Elefson	N/A	N/A
6565490	12/2002	O'Hearn	482/102	A63B 21/0628
6685601	12/2003	Knapp	N/A	N/A
6719672	12/2003	Ellis	N/A	N/A
D495380	12/2003	Ceppe	N/A	N/A
D502515	12/2004	Carter	N/A	N/A
D504164	12/2004	Ceppe	N/A	N/A
D523496	12/2005	Grider	N/A	N/A
7070546	12/2005	Grasso	482/145	A63B 21/072
7211030	12/2006	Cao	N/A	N/A
D557758	12/2006	Fedriga	N/A	N/A
7374516	12/2007	Lundquist	N/A	N/A
7488277	12/2008	Knapp	N/A	N/A
7608024	12/2008	Sechrest	N/A	N/A
7632221	12/2008	Kolander	482/102	A63B 23/03525
7641602	12/2009	Rogers et al.	N/A	N/A
7837600	12/2009	Habing	N/A	N/A
D629472	12/2009	Rhodes	N/A	N/A
7918771	12/2010	Rogers et al.	N/A	N/A
8002677	12/2010	Dibble et al.	N/A	N/A
8147390	12/2011	Rogers et al.	N/A	N/A
8257233	12/2011	Rogers et al.	N/A	N/A
8491449	12/2012	Rogers et al.	N/A	N/A
8657726	12/2013	Rogers et al.	N/A	N/A
8876675	12/2013	Califano	N/A	N/A
9011301	12/2014	Balandis	N/A	N/A
D734410	12/2014	Childs	N/A	N/A
D735820	12/2014	Alenaddaf	N/A	N/A
D735821	12/2014	Alenaddaf	N/A	N/A
D737387	12/2014	De Siqueira Indio da Costa	N/A	N/A
9192799	12/2014	Alenaddaf	N/A	N/A
D746387	12/2014	Alessandri	N/A	N/A
D755908	12/2015	James	N/A	N/A
D757189	12/2015	Colburn	N/A	N/A
9327161	12/2015	Maher	N/A	N/A
9597539	12/2016	Grider	N/A	N/A
D788861	12/2016	Campanaro	N/A	N/A
9669251	12/2016	Jones-Guinasso	N/A	N/A
D803126	12/2016	Foley	N/A	N/A
D806808	12/2017	Gregory	N/A	N/A
D808475	12/2017	Meredith	N/A	N/A
9895572	12/2017	Donato	N/A	N/A
9907991	12/2017	Leipheimer	N/A	N/A
D818547	12/2017	Meredith	N/A	N/A
9999796	12/2017	Leipheimer	N/A	N/A
D841014	12/2018	Laudadio	N/A	N/A
10226661	12/2018	Henniger	N/A	N/A
10293206	12/2018	Kamins	N/A	N/A
D852291	12/2018	Meredith	N/A	N/A
D852907	12/2018	Sergakis	N/A	N/A
D853404	12/2018	Bowman	N/A	N/A
D854637	12/2018	Adams	N/A	N/A
10486010	12/2018	Leipheimer	N/A	N/A
D871468	12/2018	Delvaux	N/A	N/A
D883405	12/2019	Dunahay	N/A	N/A
D883406	12/2019	Sergakis	N/A	N/A

10646744	12/2019	Steenkamp et al.	N/A	N/A
D888850	12/2019	Smith, Jr.	N/A	N/A
D890271	12/2019	Carmichael et al.	N/A	N/A
D890867	12/2019	Carmichael et al.	N/A	N/A
D892239	12/2019	Carmichael et al.	N/A	N/A
D893639	12/2019	Henniger	N/A	N/A
D896326	12/2019	Adams	N/A	N/A
D898136	12/2019	Carmichael et al.	N/A	N/A
D898137	12/2019	Carmichael	N/A	N/A
10792532	12/2019	Hall	N/A	N/A
10799749	12/2019	Sorin	N/A	N/A
D903793	12/2019	Henniger	N/A	N/A
10857416	12/2019	Leipheimer	N/A	N/A
10864404	12/2019	Leipheimer	N/A	N/A
10953301	12/2020	Brasch et al.	N/A	N/A
D920451	12/2020	McGrotty	N/A	N/A
11103738	12/2020	Sorin et al.	N/A	N/A
11110313	12/2020	Gore	N/A	A63B 21/0783
11117015	12/2020	Steenkamp et al.	N/A	N/A
11173337	12/2020	Jones et al.	N/A	N/A
D949262	12/2021	Horita	N/A	N/A
D954864	12/2021	Carmichael et al.	N/A	N/A
D954867	12/2021	Carmichael et al.	N/A	N/A
D955511	12/2021	Carmichael et al.	N/A	N/A
2002/0193213	12/2001	Batca et al.	N/A	N/A
2003/0017918	12/2002	Webb	N/A	N/A
2003/0115955	12/2002	Keiser	N/A	N/A
2004/0259696	12/2003	Shifferaw	N/A	N/A
2005/0202941	12/2004	Vaes	N/A	N/A
2006/0030461	12/2005	Ish, III	N/A	N/A
2006/0030462	12/2005	Ish, III	N/A	N/A
2006/0217245	12/2005	Golesh	N/A	N/A
2007/0155595	12/2006	Rogers	N/A	N/A
2007/0155597	12/2006	Hockemeyer	N/A	N/A
2008/0287270	12/2007	Carter	482/99	A63B 23/12
2008/0312051	12/2007	Manyseng	N/A	N/A
2009/0124469	12/2008	Webber	N/A	N/A
2009/0203500	12/2008	Nishimura	N/A	N/A
2009/0247376	12/2008	Merrithew	482/121	A63B 23/1263
2010/0022363	12/2009	Fenster et al.	N/A	N/A
2011/0237405	12/2010	Reyes	482/94	A63B 21/0628
2013/0035220	12/2012	Adams	N/A	N/A
2013/0296144	12/2012	Gvoich	482/102	A63B 21/062
2013/0296146	12/2012	Staten	N/A	N/A
2014/0135185	12/2013	Dalebout	N/A	N/A
2015/0159804	12/2014	Bowman	N/A	N/A
2015/0352395	12/2014	Gregory	N/A	N/A
2016/0074688	12/2015	Leipheimer	N/A	N/A
2017/0246496	12/2016	Nelson	N/A	N/A
2017/0246504	12/2016	Simmons	N/A	N/A
2018/0117382	12/2017	Johanson	N/A	N/A
2018/0353795	12/2017	Ostmeyer	N/A	N/A
2019/0134450	12/2018	Kuo	N/A	A63B 21/154
2019/0240521	12/2018	Staten	N/A	N/A
2019/0269992	12/2018	Dunahay	N/A	N/A
2019/0275363	12/2018	Jones	N/A	N/A
2019/0308069	12/2018	Sandhu	N/A	N/A
2020/0038704	12/2019	Adams	N/A	N/A
2020/0114195	12/2019	Henniger	N/A	N/A
2020/0155889	12/2019	Leipheimer	N/A	A63B 21/156
2020/0179744	12/2019	Arroyo	N/A	N/A
2020/0222745	12/2019	Leipheimer	N/A	A63B 23/03525
2021/0060374	12/2020	Slater	N/A	N/A
2021/0379438	12/2020	Sorin et al.	N/A	N/A
2022/0096893	12/2021	Steenkamp et al.	N/A	N/A
2022/0105380	12/2021	Jones et al.	N/A	N/A
2022/0314055	12/2021	Wuttke	N/A	A63B 22/203
2023/0118879	12/2022	Yang	482/138	A63B 21/0442
2024/0131387	12/2023	Yang	N/A	A63B 21/4043
2024/0226643	12/2023	Yang	N/A	A63B 21/0628

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201949544	12/2010	CN	N/A
103550906	12/2013	CN	N/A
203539969	12/2013	CN	N/A
202230408621.9	12/2021	CN	N/A
1125599	12/2000	EP	N/A
3019970012320	12/1997	KR	N/A
102282747	12/2020	KR	N/A
2006008767	12/2005	WO	N/A
2018/086028	12/2017	WO	N/A

OTHER PUBLICATIONS

REP Fitness, ARES Builder, published on Oct. 13, 2022 [online]. Retrieved from internet (Date: Sep. 12, 2023), <https://repfitness.com/collections/specialty/products/ares-cable-attachment-builder> (Year: 2022), 1 page. cited by applicant

Garage Gym Reviews, REP Fitness ARES Cable Attachment, published on Sep. 28, 2022 [online]. Retrieved from internet (Date: Sep. 12, 2023), <https://www.youtube.com/watch?v=yw2V9DdnuU> (Year: 2022), 1 page. cited by applicant

Major Lutie, Multifunction Power Cage, published on Feb. 3, 2023 [online]. Retrieved from internet (Date: Sep. 13, 2023), <https://www.amazon.com/dp/B0BVLM5B95/?th=1> (Year: 2023), 1 page. cited by applicant

Major Lutie, Multi-Function Power Rack, published on Mar. 22, 2022 [online]. Retrieved from internet (Date: Sep. 13, 2023), <https://www.amazon.com/MAJOR-LUTIE-Multi-Function-Adjustable-Attachments/dp/B09W5N8HY3/?th=1> (Year: 2023), 1 page. cited by applicant

Marcy, Machine Cage System, published on Oct. 8, 2019 [online]. Retrieved from internet (Date: Sep. 13, 2023), <https://www.amazon.com/Marcy-Multifunction-Customizable-Training-SM-4033/dp/B07YVTW8ZF/?th=1> (Year: 2019), 1 page. cited by applicant

Marcy, Stack Home Gym Multifunction, published on Apr. 13, 2015 [online]. Retrieved from internet (Date: Sep. 12, 2023), <https://www.amazon.com/dp/B00XHN4CA/> (Year: 2015), 1 page. cited by applicant

REP Fitness, ARES Attachment (6-Post-Pre-Selected), published on Oct. 13, 2022 [online]. Retrieved from internet (Date: Sep. 13, 2023), <https://repfitness.com/collections/specialty/products/ares-6-post-pre-selected> (Year: 2022), 2 pages. cited by applicant

Spart, Functional Trainer Cable Machine, published on May 24, 2023 [online]. Retrieved from internet (Date: Sep. 12, 2023), <https://www.amazon.com/dp/B0C66CYYCQ/?th=1> (Year: 2023), 1 page. cited by applicant

Xmark, Functional Trainer Cable Machine, published on May 10, 2012 [online]. Retrieved from internet (Date: Sep. 12, 2023), <https://www.amazon.com/dp/B008230QHU/?th=1> (Year: 2012), 1 page. cited by applicant

Belt Squat by Rep Fitness dated Jul. 9, 2020. Found online [Jan. 7, 2021] <https://www.youtube.com/watch?v=9wn6IYVKIXY> (Year: 2020). cited by applicant

The Fitness Outlet, BodyCraft RFT Rack Functional Trainer, located online on Mar. 6, 2023 at: https://www.thefitnessoutlet.com/bodycraft-rft-rack-functional-trainer/?utm_source=google&utm_medium=cse&utm_term=5090&gclid=CjwKCAiA5sieBhBnEiWAR9oh2kPf6ZzAr_PHj9zMpMT5x-DmhUOizW0O2LvypV9TtEiv7j9gus9PROCqF8QAvD_BWE, offered for sale as early as Jan. 7, 2022, 6 pages. cited by applicant

Internet Archive Wayback Machine, BodyCraft RFT Rack Functional Trainer, for link <https://www.bodycraft.com/product/f430-f438-rft>; located online on Nov. 28, 2020 at: <http://web.archive.org/web/20210121031323/https://www.bodycraft.com/product/f430-f438-rft>, 3 pages. cited by applicant

Internet Archive Wayback Machine, Rogue LP-2 Lat Pulldown Low Row, for link <https://www.roguefitness.com/rogue-p-2-lat-pulldown-low-row>, located at <http://web.archive.org/web/20200812230501/https://www.roguefitness.com/rogue-p-2-lat-pulldown-low-row>, dated Aug. 12, 2020, 4 pages. cited by applicant

Internet Archive Wayback Machine, Prime HLP Selectorized Rack, for link <https://primefitnessusa.com/hlp-selectorized-rack/>, located at: <http://web.archive.org/web/20210123003723/https://primefitnessusa.com/hlp-selectorized-rack/>, dated Aug. 14, 2020, 3 pages. cited by applicant

European Extended Search Report in Application 23150155.2, mailed May 31, 2023, 7 pages. cited by applicant

Vendetta Strength and Athletics, Titan Lever Arm—180 degree Adapter Mounts (Pair), located online at: <https://vendettastrengthandathletics.com/products/adjustable-lever-arm-titan-adapter-mounts-pair>, captured on WayBack Internet Archive on Oct. 20, 2022, 5 pages. cited by applicant

Garage Gym Outfitters, Crandall Fitness Adjustable Free Motion Swing Arms, located online on Mar. 6, 2023 at: <https://thegaragegymoutfitters.com/products/crandall-fitness-adjustable-free-motion-swing-arms>, 5 pages. cited by applicant

Griffin Fitness, Griffin Adjustable Lever Arms, located online at: <https://griffin.fitness/rack-attachments/griffin-adjustable-lever-arms/>, captured on WayBack internet Archive on Feb. 10, 2022, 4 pages. cited by applicant

Rep Fitness, ISO Arms, located online on Mar. 6, 2023 at: https://repfitness.com/products/iso-arms?psafe_param=1&gclid=CjwKCAjw8symBhAqEiwAaTA_HEoleFUX_hO9PcY99fM57nLRuNt7bRws9hJAIQs_Bmb74I0F3RatRoCe2QQAvD_Bw, 6 pages. cited by applicant

USA Home Gym, Inspire Fitness SCS Smith Cage Sytem Review, published Aug. 4, 2014, <https://www.USA-homegym.com/smith-machine/inspire-fitness-scs-smith-cage-system/>, printed on Oct. 13, 2023, 6 pages. cited by applicant

Flame Sport Professional Gym Equipment, Multifunctional Power Rack (9B), published Jul. 12, 2017 on Wayback Machine, <https://www.flame-sport.de/en/multifunctional-power-rack-9b.html>, printed on Oct. 13, 2023, 5 pages. cited by applicant

Danboarding, You Tube video, FIT505 Universal Functional Smith Machine, published on Oct. 7, 2017, <https://www.youtube.com/watch?v=gpkov9GopzA>, duration 6:43, 6 pages. cited by applicant

BRUTEforce Australia, You Tube video, Exercise Guide—BRUTEforce 360PTA Functional Trainer—Renouf Fitness, published Oct. 20, 2017, <https://www.youtube.com/watch?v=fVBu7j6uZkc>, duration 7:54, 1 page. cited by applicant

European patent application No. 23150155.2, Communcation Pursuant to Article 94(3) EPC dated Jan. 15, 2025. 4 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Application No. 63/297,300, filed Jan. 7, 2022, which is hereby incorporated in its entirety by reference.

INTRODUCTION

(1) A variety of machines exist to train specific muscles or muscle groups. For example, leg curl machines may be used to exercise the hamstrings, while leg extension machines may be used to exercise the quadriceps. For the upper body, pull-down machines are used to exercise the latissimus dorsi, while so-called “pec decks” may be used to exercise the pectorals. Other specific machines for specific body parts are known. Functional trainers, however, may be used to exercise multiple body parts, but due to their versatility, these machines are often exceedingly large. Further, due to the complexity thereof, functional trainers are often very expensive pieces of exercise equipment, making them cost-prohibitive for home gym users or even commercial gyms with limited budgets.

SUMMARY

(2) In one aspect, the technology relates to an exercise system including: a base structure including: a first cross brace including first cross brace mounting plates disposed at each end of the first cross brace; a pair of base guide rail supports secured to and extending substantially orthogonal from the first cross brace, wherein each of the pair of base guide rail supports includes a pair of base guide rail receivers; a pair of base pulley supports secured to and extending substantially orthogonal from the first cross brace, wherein each of the pair of base pulley supports further includes: a pulley-support pulley housing; and a pair of pulley-support pulleys rotatably secured to the pulley-support pulley housing; and a second cross brace including: a central portion substantially parallel to the first cross brace, wherein the central portion is secured to the pair of base guide rail supports and the pair of base pulley supports; two end portions disposed at an angle to the central portion, wherein each of the two end portions terminate at second cross brace mounting plates disposed at each end of the two end portions; two central-portion pulley housings; and a central-portion pulley rotatably secured to each of the two central-portion pulley housings. In an example, the exercise system further includes a pair of weight stack units, wherein each of the pair of weight stack units includes: a pair of weight stack guide rails, wherein each of the pair of weight stack guide rails are secured at lower ends to one of the pair of base guide rail receivers on a same one of the pair of base guide rail supports; a movable weight stack slidably engaged with the pair of weight stack guide rails; and a lifting rod selectively engageable with the movable weight stack. In another example, the exercise system further includes an upper structure including: a third cross brace including third cross brace mounting plates disposed at each end of the third cross brace; a pair of upper guide rail supports secured to and extending substantially orthogonal from the third cross brace, wherein each of the pair of upper guide rail supports includes a pair of upper guide rail receivers, and wherein each of the pair of weight stack guide rails are secured at upper ends to one of the pair of upper guide rail receivers on a same one of the pair of upper guide rail supports; a pulley support plate fixed to each of the pair of upper guide rail supports, wherein each pulley support plate includes a pair of pulley-support plate pulley housings and a pulley-support plate pulley rotatably secured to each of the pair of pulley-support plate pulley housings; and a fourth cross brace including: two end plates, wherein each of the two end plates is secured to a single one of the pair of upper guide rail supports; a spanner connected to each of the two end plates; two spanner pulley housings secured to the spanner; and a pulley rotatably secured to each of the two spanner pulley housings. In yet another example, the base structure further includes a foot plate slidably secured to the central portion of the base structure. In still another example, the exercise system further includes a pair of upper outer pulley bracket units, each including a bracket, a pair of upper outer pulley housings secured to the bracket, and an upper outer pulley rotatably secured to each of the pair of upper outer pulley bracket units.

(3) In another example of the above aspect, the exercise system further includes a pair of lower outer pulley bracket units, each including a bracket, three lower outer pulley housings secured to the bracket, and a lower outer pulley rotatably secured to each of the three of the lower outer pulley housings. In an example, the exercise system further includes a floating pulley unit disposed between each of the upper outer pulley bracket units and lower outer pulley bracket units, wherein each floating pulley unit includes a pair of floating pulleys rotatably coupled to a floating pulley housing. In another example, the exercise system further includes a pair of front upper pulley brackets, wherein each of the front upper pulley bracket units includes a bracket, a housing secured to the bracket, and a front upper pulley rotatably secured to the housing; and a pair of front lower pulley brackets, wherein each of the front lower pulley bracket units includes a bracket, a housing secured to the bracket, and a front lower pulley rotatably secured to the housing. In yet another example, the exercise system further includes a trolley disposed between each of the pair of front upper pulley bracket units and each of the pair of front lower pulley bracket units, wherein each trolley includes: a receiver; a pin coupled to the receiver for selectively securing the receiver to a frame; a handle secured to the receiver; and a roller rotatably secured to the receiver. In still another example, the exercise system further includes a cable for rotatably coupling any two of the plurality of pulleys.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawing figures, which form a part of this application, are illustrative of described technology and are not meant to limit the scope of the disclosure as claimed in any manner, which scope shall be based on the claims appended hereto.

(2) FIG. 1 depicts a perspective view of a weightlifting system, mounted to a frame.

(3) FIG. 2 depicts a perspective view of the weightlifting system of FIG. 1, without the frame.

(4) FIG. 3 depicts an enlarged upper perspective view of an upper structure of the weightlifting system of FIG. 2.

(5) FIG. 4 depicts an enlarged lower perspective view of the upper structure of FIG. 3.

(6) FIG. 5 depicts a partially exploded, enlarged upper perspective view of a base structure of the weightlifting system of FIG. 2.

(7) FIG. 6 depicts an enlarged upper perspective view of a trolley and a front lower pulley bracket unit of the weightlifting system of FIG. 2.

(8) FIG. 7 depicts a partial enlarged upper perspective view of a base structure and a lower outer pulley bracket unit of the weightlifting system of FIG. 2.

(9) FIG. 8 depicts a partial enlarged upper perspective view of an upper structure, upper outer pulley bracket unit, and a floating pulley unit of the weightlifting system of FIG. 2.

(10) FIG. 9 depicts an enlarged upper perspective view of a weight system of the weightlifting system of FIG. 2.

(11) FIG. 10 depicts an enlarged perspective view of a front upper pulley bracket unit of the weightlifting system of FIG. 2.

(12) FIG. 11 depicts a schematic drawing of a cable routing configuration for the weightlifting system of FIG. 2.

DETAILED DESCRIPTION

(13) FIGS. 1-5 depict various views of a weightlifting system **100**, which may also be referred to herein as a functional trainer or exercise system, as well as components thereof. The figures are described concurrently and not all components of the weightlifting system **100** are depicted or labeled in every figure. In FIG. 1, a frame structure **Fr** is depicted. The frame structure **Fr** may be formed of steel or other robust metal tubing sized as required or desired for a particular application, e.g., 1-½", 2", 3" and larger. Tubing formed of 12G, 10G, 8G, and larger gauge materials are contemplated. The steel tubing of the frame is perforated at regular intervals with openings sized to receive bolts, pins, screws, or other elongate elements that project from the various components that make up the weightlifting system **100**. For clarity, the components of the weightlifting system **100** that are described elsewhere herein are depicted in solid line.

(14) The various components of the weightlifting system **100** may be attached to the frame **Fr** or other components of the weightlifting system **100** with one or more of mounting plates, brackets, or receivers. As used herein, the term "mounting plate" refers to a flat plate or multiple plates defining one or more openings for receiving a bolt, pin, or other elongate element that may be used to secure the mounting plate to the frame **Fr**. In examples, mounting plates are attached to ends of components and are secured via bolts to adjacent components or the frame **Fr**; in other examples, mounting plates may be welded to adjacent components or the frame **Fr**. Brackets are similar to mounting plates and may include one or more flat plates to secure components to adjacent components or the frame **Fr**. In other examples, brackets may be welded to other components or the frame **Fr**. Finally, in examples, the term "receiver" contemplates two or more flat plates into which a portion of the frame **Fr** are received or inserted. Certain of the mounting plates, brackets, or receivers may be selectively movable and/or positionable. In examples, it may be advantageous to utilize bolts or pins or other removable elements to secure the mounting plates, brackets, and receivers to the frame **Fr** or other components of the weightlifting system **100**. Use of mechanical fasteners may enable an end user to install the weightlifting system on their own frame **Fr**. Welded components may typically be more appropriate for components shipped fully assembled from a manufacturer. Regardless, the type of connection elements (mounting plate, bracket, receiver, etc.) and the type of connection to an adjacent element (bolt, weld, pin, etc.) should not be considered limiting. The functional trainer described herein is depicted for illustrative purposes only for certain contemplated configurations; other configurations would be apparent to a person of skill in the art.

(15) The exercise or weightlifting system **100** includes a base structure **200**, a weight system **300** including pair of weight stacks, and an upper structure **400**, along with a number of pulleys located remote therefrom. The base structure **200** includes a first cross brace **202** having first cross brace mounting plates **204** disposed at each end of the first cross brace **202**. These mounting plates **204** are secured to the frame **Fr**, with bolts or other fasteners (not shown). A pair of base guide rail supports **206** are secured to and extending substantially orthogonal from the first cross brace **202**. Each of the pair of base guide rail supports **206** define a pair of base guide rail receivers **208**. The base guide rail receivers **208** may be openings in the base guide rail supports **206**, or may be a separate structure (e.g., tubing, plate, or other structure) that may receive a bottom end of the guide rails of the weight stacks (described in more detail below). A pair of base pulley supports **210** are secured to and extend substantially orthogonal from the first cross brace **202**. As such, each of the base pulley supports **210** are substantially parallel to the base guide rail supports **206**. Each of the pair of base pulley supports **210** include a pulley-support pulley housing **212**, which extends from the base pulley support **210**, generally away from an axis **X**, as depicted, e.g., in FIG. 5. A pair of pulley-support pulleys **214** are rotatably secured to the pulley-support pulley housing **212**, generally about axles that are skew relative to each other. The pulley-support pulleys **214** allow for redirection of a cable (not depicted in this figure), as do the other pulleys in weightlifting system **100**.

(16) Secured to opposite ends of the base pulley supports **210** and the base guide rail supports **206** is a second cross brace **216**. The second cross brace **216** includes a central portion **218** that is substantially parallel to the first cross brace **202**. The central portion **218** is secured to the pair of base guide rail supports **206** and the pair of base pulley supports **210**. Further, two central-portion pulley housings **220** each have rotatably attached thereto a central-portion pulley **222**. The second cross brace **216** also includes two end portions **224** disposed at an angle to the central portion **218**. In examples, the angle may be about 105°, about 110° about 115°, about 120°, about 125°, and about 130°. In examples, the angle may be between about 110° and about 130°, or between about 115° and about 125°. In other examples, the angle may be about 121°, about 122°, about 123°, and about 124°. The angled offset from the central portion **218** allows the major components of the weightlifting system **100** to be more compact, and to be set farther back within the frame **Fr** depicted in FIG. 1, while still being attachable to a vertical portion of the frame **Fr** at second cross brace mounting plates **226** disposed at each end of the two end portions **224**. A receiver **228** may be configured to receive a post **230** extending from a foot plate **232**, which may enable additional exercises with the weightlifting system **100**. A distance between the central portion **218** and the foot plate **232** may be adjustable by selective operation of an adjustment pin **234**.

(17) Resistance to exercises performed with the weightlifting system **100** is provided by the weight system **300**. The depicted weight system **300** includes two movable weight stack units **302**. The weight stack units **302** may include selectively removable pins (not shown) for selecting a desired weight for a particular exercise. The pin projects into a lifting rod **304** and a lifting force of a cable around a weight stack pulley **306** lifts the selected portion of the weight stack unit **302** during exercise. When lifted and lowered, each weight stack unit **302** slides along a pair of weight stack guide rails **308**. Each of the weight stack guide rails **308** are secured at lower ends to one of the pair of base guide rail receivers **208** on a same one of the pair of base guide rail supports **206**. One or more shrouds **310** may substantially surround the weight stack units **302** to prevent inadvertent contact between a user and the moving weight stacks units **302** during movement.

(18) The weightlifting system **100** also includes an upper structure **400**, which includes a third cross brace **402**. Third cross brace mounting plates **404** are disposed at each end of the third cross brace **402**, for securing the third cross brace **402** to an upright or other portion of the frame **Fr**. A pair of upper guide rail supports **406** are secured to and extend substantially orthogonal from the third cross brace **402**. In the depicted example, the upper guide rail supports **406** are secured to the third cross brace **402** with mounting plates **408**. Each of the pair of upper guide rail supports **406** include a pair of upper guide rail receivers **410**, which may be similar in configuration to the base guide rail receivers **208**, into which each of the pair of weight stack guide rails **308** are secured at upper ends thereof, e.g., on a same one of the pair of upper guide rail supports **406**. A pulley support plate or bracket **412** is fixed to each of the pair of upper guide rail supports **406**. Each pulley support plate **412** includes a pair of pulley-support plate pulley housings **414**. Each pulley support housing **414** has rotatably secured thereto a pulley-support plate pulley **416**. In examples, the pairs of pulley support housings **414** on a single one of the upper guide rail supports **406** may be constructed as a single housing, such as the pulley-support pulley housing **212**, depicted above. The upper structure **400** further includes a fourth cross brace **418** that, in this example, is constructed of a number of parts. Here, two end plates **420** are secured to individual ends of each of the upper guide rail supports **406**. These two end plates **420** are joined via a spanner **422**. The spanner **422** is configured as a bracket to substantially surround a portion of tubing that forms a part of the frame **Fr**. Two spanner pulley housings **424** are secured to the spanner **422** and each include a spanner pulley **426** rotatably secured thereto.

(19) The base structure **200**, weight system **300**, and upper structure **400** form the majority of the components of the weightlifting system **100**. A plurality of pulleys are also utilized to guide a plurality of cables to enable a wide range of range exercises. A pair of upper outer pulley bracket units **500** are disposed proximate and on each side of the upper structure **400**. Each upper outer pulley bracket unit **500** includes a bracket **502**, and a pair of upper outer pulley housings **504** secured to the bracket **502**. In examples, the pairs of upper outer pulley housings **504** on a single one of the brackets **502** may be constructed as a single housing, such as the pulley-support pulley housing **212**, depicted above. Regardless, two upper outer pulleys **506** are rotatably secured to each of the brackets **502**, via one or more housings **504**. Disposed below each upper outer pulley bracket unit **500** is a floating pulley unit **520**. Each floating pulley unit **520** includes a pair of floating pulleys **522** rotatably coupled to a floating pulley housing **524**. Disposed below each floating pulley unit **520** is a lower outer pulley bracket unit **540**, each of which includes a bracket **542**. Depending on configuration, one, two or three lower outer pulley housings **544** may be secured to the bracket **542**. A lower outer pulley **546** is rotatably secured to each of the brackets **542**, via the lower outer pulley housings **544**.

(20) Proximate a front of the frame **Fr** are a pair of front upper pulley bracket units **560** and a pair of front lower pulley bracket units **580**. Each of the front upper pulley bracket units **560** includes a bracket **562** and a housing **564** secured to the bracket **562**. A front upper pulley **566** is rotatably secured to the housing **564**. Similarly, each of the front lower pulley bracket units **580** includes a bracket **582** and a housing **584** secured to the bracket **582**. A front lower pulley **586** is rotatably secured to the housing **584**. Disposed between each of the pair of front upper pulley bracket units **560** and each of the pair of front lower pulley bracket units **580** is a trolley **600**, which is positionable vertically along from the frame **Fr**, in examples, as high as the front upper pulley bracket unit **560** and as low as front lower pulley bracket unit **580**. Each trolley **600** includes a receiver **602** that is positionable along a portion of the frame **Fr** to which it is mounted. A pin **604** is coupled to the receiver **602** for selectively securing the receiver **602** to the frame **Fr**. A handle **606** is secured to the receiver **602** to aid in allowing a user of the weightlifting system **100** in positioning the trolley **600** vertically between the front upper pulley bracket units **560** and the front lower pulley bracket units **580**. At least one roller **608** is rotatably secured to the receiver **602**, e.g., via a housing **610**. In examples, multiple rollers **608** may be rotatably secured to each housing **610**. A cable anchor **612** projects from the receiver **602**.

(21) FIGS. **6-10** depict, in relevant part, enlarged views of various pulleys of the weightlifting system of FIG. **2**. The numbering used in FIGS. **6-10** is consistent with that used in FIGS. **1-5**, except specific pulleys and rollers are designated by letters A-R. FIGS. **6-10** depict pulleys and rollers A-R, as described in the context of the cable routing configuration **700** depicted in FIG. **11**. FIG. **11** depicts a schematic cable routing configuration **700** on a single side of the weightlifting system **100**, connecting pulleys A-R. A mirror image of the cable routing configuration **700** is present on the opposite side of the weightlifting system **100**. Only certain other components are depicted and described. Further, the cable routing configuration **700** is for illustrative purposes only; that is, a person of ordinary skill in art, upon reading this disclosure, would understand whether a cable wraps around a certain perimeter portion of a particular pulley or roller, based on the entrance position and exit position of the cable, relative to the pulley or roller. The cable routing configuration **700** includes a lower cable **LC** and an upper cable **UC**.

(22) Lower cable **LC** is anchored at one end to anchor **612**, which is secured to trolley housing **610**. It then is passed around pulleys A, B, and C, before being passed around pulley D, in the floating pulley unit housing **524**, which also includes therein pulley M. Lower cable **LC** is then passed around pulleys E, F, G, and H. After pulley H, a second end of lower cable **LC** may be connected to a first handle **702**. A second handle **704** is at a first end of upper cable **UC**. Upper cable passes between pulleys I and J in trolley housing **610**, then passes around pulleys K and L, before being passed around pulley M, in the floating pulley unit housing **524**. Upper cable **UC** is then passed around pulleys N and O, before passing around pulley P, the actuation of which raises and lowers the weight stack unit **302**. Thereafter, the upper cable passes around pulleys Q and R. A second end of upper cable **UC** is connected to a third handle **706**. Since the lower cable **LC** is anchored at an anchor **612** to the trolley housing **610**, movement of the trolley **600** up or down on the frame **Fr** causes a corresponding movement of the floating pulley unit **520**, to take up the slack of either cable **LC**, **UC**. Pulling of any handle **702**, **704**, **706** moves the weight stack unit **302** up, providing resistance to a user of the weightlifting system **100**.

(23) It is to be understood that this disclosure is not limited to the particular structures, process steps, or materials disclosed herein, but is extended to equivalents thereof as would be recognized by those ordinarily skilled in the relevant arts. It should also be understood that terminology employed herein is used for the purpose of describing particular examples only and is not intended to be limiting. It must be noted that, as used in this specification, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise.

(24) It will be clear that the systems and methods described herein are well adapted to attain the ends and advantages mentioned as well as those inherent therein. Those skilled in the art will recognize that the methods and systems within this specification may be implemented in many manners and as such is not to be limited by the foregoing exemplified examples and examples. In this regard, any number of the features of the different examples described herein may be combined into one single example and alternate examples having fewer than or more than all of the features herein described are possible.

(25) While various examples have been described for purposes of this disclosure, various changes and modifications may be made which are well within the scope contemplated by the present disclosure. Numerous other changes may be made which will readily suggest themselves to those skilled in the art and which are encompassed in the spirit of the disclosure.

Claims

1. An exercise system comprising: a base structure comprising: a first cross brace comprising first cross brace mounting plates disposed at each end of the first cross brace; a pair of base guide rail supports secured to and extending substantially orthogonal from the first cross brace, wherein each of the pair of base guide rail supports comprise a pair of base guide rail receivers; a pair of base pulley supports secured to and extending substantially orthogonal from the first cross brace, wherein each of the pair of base pulley supports further comprises: a pulley-support pulley housing; and a pair of pulley-support pulleys rotatably secured to the pulley-support pulley housing; and a second cross brace comprising: a central portion substantially parallel to the first cross brace, wherein the central portion is secured to the pair of base guide rail supports and the pair of base pulley supports; two end portions disposed at an angle to the central portion, wherein each of the two end portions terminate at second cross brace mounting plates disposed at each end of the two end portions; two central-portion pulley housings; and two central-portion pulleys each rotatably secured to a respective one of the two central-portion pulley housings; and a pair of weight stack units configured to seat upon the pair of base guide rail supports, respectively, wherein each of the pair of weight stack units comprises: a pair of weight stack guide rails, wherein each of the pair of weight stack guide rails are secured at lower ends to a respective one of the pair of base guide rail receivers on a same one of the pair of base guide rail supports; a movable weight stack slidably engaged with the pair of weight stack guide rails; a lifting rod selectively engageable with the movable weight stack; and a weight stack

pulley connected to the lifting rod; wherein the first cross brace, the pair of base guide rail supports, the pair of base pulley supports, and the second cross brace each reside within a first single plane.

2. The exercise system of claim 1, further comprising: an upper structure comprising: a third cross brace comprising third cross brace mounting plates disposed at each end of the third cross brace; a pair of upper guide rail supports secured to and extending substantially orthogonal from the third cross brace, wherein each of the pair of upper guide rail supports comprise a pair of upper guide rail receivers, and wherein each of the pair of weight stack guide rails are secured at upper ends to a respective one of the pair of upper guide rail receivers on a same one of the pair of upper guide rail supports; two pulley support plates each fixed to a respective one of the pair of upper guide rail supports, wherein each pulley support plate comprises a pair of pulley-support plate pulley housings and a pair of pulley-support plate pulleys, wherein each pulley-support plate pulley of each of the pairs of pulley-support plate pulleys is rotatably secured to a respective pulley-support plate pulley housing of each of the pairs of pulley-support plate pulley housings; and a fourth cross brace comprising: two end plates, wherein each of the two end plates is secured to a single respective one of the pair of upper guide rail supports; a spanner connected to each of the two end plates; two spanner pulley housings secured to the spanner; and a respective spanner pulley rotatably secured to each of the two spanner pulley housings.

3. The exercise system of claim 2, further comprising a frame connected to and extending between each of the base structure and the upper structure; a pair of upper outer pulley bracket units connected to the frame, each comprising a bracket, a pair of upper outer pulley housings secured to the bracket, and a pair of upper outer pulleys, wherein each upper outer pulley of each of the pairs of upper outer pulleys is rotatably secured to a respective upper outer pulley housing of each of the pairs of upper outer pulley housings.

4. The exercise system of claim 3, further comprising a pair of lower outer pulley bracket units connected to the frame, each comprising a bracket, three lower outer pulley housings secured to the bracket, and three lower outer pulleys, wherein each of the three lower outer pulleys of each of the pair of lower outer pulley bracket units is rotatably secured to a respective one of the three lower outer pulley housings.

5. The exercise system of claim 4, further comprising two floating pulley units each disposed between a respective upper outer pulley bracket unit of the pair of upper outer pulley bracket units and a respective lower outer pulley bracket unit of the pair of lower outer pulley bracket units, wherein each floating pulley unit comprises a pair of floating pulleys rotatably coupled to a floating pulley housing.

6. The exercise system of claim 5, further comprising: a pair of front upper pulley bracket units attached to upper portions of the frame, wherein each of the pair of front upper pulley bracket units comprises a bracket, a housing secured to the bracket, and a front upper pulley rotatably secured to the housing; and a pair of front lower pulley bracket units attached to lower portions of the frame beneath a respective front upper pulley bracket unit, wherein each of the front lower pulley bracket units comprises a bracket, a housing secured to the bracket, and a front lower pulley rotatably secured to the housing.

7. The exercise system of claim 6, further comprising: two trolleys, each disposed between a respective front upper pulley bracket unit of the pair of front upper pulley bracket units and a respective front lower pulley bracket unit of the pair of front lower pulley bracket units, wherein each trolley comprises: a receiver; a pin coupled to the receiver for selectively securing the receiver to the frame; a handle secured to the receiver; and a roller rotatably secured to the receiver.

8. The exercise system of claim 7 further comprising: a first pair of cables moveably terminating at respective first ends at respective ones of the two trolleys, moveably terminating at respective second ends at respective ones of the two spanner pulleys, and traveling around respective ones of the front upper pulleys, upper pulleys in the pairs of floating pulleys, the single floating pulley, and the pulley support-plate pulleys; and a second pair of cables fixedly terminating at respective first ends at respective ones of the two trolleys, moveably terminating at respective second ends at respective ones of the two central portion pulleys, and traveling around respective ones of the front lower pulleys, lower pulleys in the pairs of floating pulleys, the pair of pulley-support pulleys, and the two central portion pulleys.

9. The exercise system of claim 6, further comprising a cable for rotatably coupling any two pulleys between one each of the front upper pulleys, the front lower pulleys, the pairs of floating pulleys, the three lower outer pulleys, the pairs of upper outer pulleys, the pair of pulley-support plate pulleys, the spanner pulleys, the two central-portion pulleys, or the pair of pulley-support pulleys.

10. The exercise system of claim 4, wherein a first rotational axis of one of the three pulleys in each lower pulley bracket unit is orthogonal to second and third rotational axes of the others of the three pulleys, respectively.

11. The exercise system of claim 2, wherein the third cross brace, the fourth cross brace, and the pair of upper guide rail supports of the upper structure each reside within a single second plane.

12. The exercise system of claim 1, wherein the base structure further comprises a foot plate slidably secured to the central portion of the second cross brace of the base structure.
